

Normal Ideals

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1 Regularity Properties

1.1 Precipitousness

- I. An ideal I on a regular cardinal κ is always assumed to be uniform¹.
- II. **Definition.** The poset $\mathbb{P}_I$ is defined as the set $I^+ = P(\kappa) - I$, ordered by the inclusion.
- III. **Proposition.** Let $G \rightsquigarrow V^{\mathbb{P}_I}$.
 - a. G is a V -ultrafilter on κ extending the filter I^* dual to I .
 - b. If I is κ -complete, then G is V - κ -complete.
 - c. If I is normal, then G is V -normal.

Proof.

Part a.

1° Routine.

Part b.

¹This means that for all $X \subseteq \kappa$ with $|X| < \kappa$, we have $X \in I$.

2° Let $\lambda < \kappa$ and let $f \in G^\lambda \cap V$.

3° $X := \bigcap_{\xi < \lambda} f(\xi) \in I^+$

Proof.

1' Assume otherwise, i.e. $X \in I$.

2' We want to show that $\{f(\xi)^{\mathbb{G}} : \xi < \lambda\}$ is predense.

3' Let $Z \in I^+$ be arbitrary.

4' We have that

$$\left(\bigcup_{\xi < \lambda} (Z \cap f(\xi)^{\mathbb{G}}) \right) / I = Z / I > 0.$$

5' By the κ -completeness of I , we conclude that there is $\xi < \lambda$ such that $(Z \cap f(\xi)^{\mathbb{G}}) / I > 0$.

6' This shows the pre-density.

7' This implies that there is a $\xi < \lambda$ with $f(\xi)^{\mathbb{G}} \in G$ - a contradiction. $\square$

4° We now have that $\{X\} \cup \{f(\xi)^{\mathbb{G}} : \xi < \lambda\}$ is predense in I^+ . This is shown analogously to 3°2'.

5° Since $G \cap \{f(\xi)^{\mathbb{G}} : \xi < \lambda\} = \emptyset$, we conclude $X \in G$.

Part c.

6° Let $S \in G$ and let $f : S \rightarrow \kappa$ with $f \in V$ and

$$(\forall \alpha \in S)(f(\alpha) < \alpha).$$

7° **Claim.** $D := \{T \in \mathbb{P}_I : T \subseteq S \wedge f \restriction T = \text{const.}\}$ is dense below S .

Proof.

1' Let $T \in \mathbb{P}_I^V : T \subseteq S$ be arbitrary.

2' Then $f \restriction T$ satisfies

$$(\forall \alpha \in T)(f(\alpha) < \alpha).$$

3' This implies existence of a $T' \in \mathbb{P}_I^V$ such that $f \restriction T' = \text{const.}$

4' We now have $T' \in D$ and $T' \subseteq T$. $\square$

8° This claim and the fact that $S \in G$ implies that

$$\exists T \in G \cap D.$$

9° In other words, $T \in G$ with $T \subseteq S$ and $f \restriction T = \text{const.}$ $\square$

IV. In particular, we can form the ultrapower $\text{Ult}(V, G)$.

V. **Proposition.** Suppose κ is a regular cardinal, I is an ideal on κ , and G is V -generic for $\mathbb{P}_I$. Then the following holds.

a. If I is κ -complete in V , then

$$\text{crit } j_G = \kappa \text{ and } \kappa \subseteq \text{wfp Ult}(V, G).$$

b. If I is normal in V , then

$$\kappa = [\text{id}] \text{ and } \kappa \in \text{wfp Ult}(V, G).$$

Proof.

1° Let $j := j_G$.

2° For **a**, we need to establish $\text{crit } j = \kappa$.

3° First we show that $\text{im } (j \restriction \kappa)$ is an initial segment of $\text{Ord}^{\text{Ult}(V, G)}$, which will have for a consequence that $j \restriction \kappa = \text{id}$.

4° Let $\alpha < \kappa$ and $[f] < j(\alpha)$ be arbitrary, with $f \in (\kappa \rightarrow V) \cap V$.

5° $\{\xi < \kappa : f(\xi) < \alpha\} \in G$.

6° Since G is V - κ -complete V -ultrafilter, there is $\beta < \alpha$ such that

$$\{\xi < \kappa : f(\xi) = \beta\} \in G.$$

I.e. $[f] = j(\beta)$.

7° To conclude **a**, it now suffices to note $j(\kappa) > [\text{id}] > j(\alpha)$ for all $\alpha < \kappa$.

8° For **b**, it suffices to show that for all $[f] < [\text{id}]$, there is $\alpha < \kappa$ such that $[f] = j(\alpha)$; but this is simply V -normality of G .

□

VI. **Definition.** An ideal I on κ is precipitous if for every $G \rightsquigarrow V^{\mathbb{P}_I}$, ultrapower $\text{Ult}(V, G)$ is well-founded.

VII. **Definition.** Given an ideal I on κ , we define *the precipitousness game* $\text{PrecGame}(I)$ as follows. A play is of the form

$$\begin{array}{c|cccc} \text{I} & S_0 & & S_2 & \cdots \\ \hline \text{II} & & S_1 & & S_3 & \cdots \end{array}$$

where $S_i \in I^+$ and $S_i \supseteq S_{i+1}$ for $i < \omega$. Player I wins if $\bigcap_{i < \omega} S_i = \emptyset$, otherwise II wins.

VIII. **Proposition.** Ideal I is precipitous if and only if player I does not have a winning strategy in $\text{PrecGame}(I)$.

Proof.

Implication ($\Leftarrow$)

- 1° Assume that I is not precipitous and let us describe a winning strategy σ for I. While producing the strategy, we also produce auxiliary functions $(f_{2n} : n < \omega)$ (which depend on the play).
- 2° Let $S_0 \in \mathbb{P}_I$ and $f_0 : \kappa \rightarrow \mathbf{Ord}$ be such that $S_0 \Vdash \ulcorner \mathbf{Ult}(V, \mathbf{G}) \urcorner$ is ill-founded below $[f_0]^\ulcorner$. We set $\sigma(\emptyset) := S_0$.
- 3° Suppose that moves $(S_i : i < 2n + 2)$ have been played and that functions $(f_{2i} : i < n + 1)$ have been produced so to satisfy $f_{2i} : \kappa \rightarrow \mathbf{Ord}$ and $S_{2i} \Vdash \ulcorner \mathbf{Ult}(V, \mathbf{G}) \urcorner$ is ill-founded below $[f_{2i}]^\ulcorner$ ($i < n + 1$).
- 4° Let $S'_{2n+2} \in \mathbb{P}_I$ with $S'_{2n+2} \subseteq S_{2n+1}$ and $f_{2n+2} : \kappa \rightarrow \mathbf{Ord}$ be such that $S'_{2n+2} \Vdash [f_{2n+2}] < [f_{2n}] \wedge \ulcorner \mathbf{Ult}(V, \mathbf{G}) \urcorner$ is ill-founded below $[f_{2n+2}]^\ulcorner$.
- 5° We set $\sigma((S_i : i < 2n + 2)) = S_{2n+2} := \{\xi \in S'_{2n+2} : f_{2n+2}(\xi) < f_{2n}(\xi)\} \in \mathbb{P}_I$.
- 6° This concludes the construction.
- 7° A play $(S_n : n < \omega)$ according to σ is won by I: otherwise, there would be $\xi \in \bigcap_{n < \omega} S_n$, witnessing $f_{2n+2}(\xi) < f_{2n}(\xi)$ for every $n < \omega$.

Implication ($\Rightarrow$)

- 8° Suppose that I is precipitous and assume towards contradiction that there is a winning strategy σ for I.
- 9° Let poset $\mathbb{P}$ be the tree of all partial plays of $\mathbf{PrecGame}(I)$ according to σ and let g be a V -generic branch for $\mathbb{P}$. We work in $V[g]$.
- 10° Let G be the filter on $\mathbb{P}_I^V$ generated by $\{g(n) : n < \omega\}$.
- 11° **Claim.** $G \rightsquigarrow V^{\mathbb{P}_I}$

Proof.

- 1' Let $D \subseteq \mathbb{P}_I$ be an arbitrary dense set.
- 2' Let $D' := \{p \in \mathbb{P} : (\exists i < |p|)(p(i) \in D)\}$. We want to show that D' is dense in $\mathbb{P}$.
- 3' Let $p_0 \in \mathbb{P}$ be arbitrary. We want to find $p \in D'$ with $p \leq p_0$.
- 4' If $|p_0|$ is even, set $S := \sigma(p)$ and choose $T \subseteq S$ such that $S \in D$. Then $p := p_0 \widehat{\ } S \widehat{\ } T$ is as required.
- 5' If $|p_0|$ is odd, choose $S \in D$ such that $S \subseteq p_0(|p_0| - 1)$. Then $p := p_0 \widehat{\ } S$ is as required.
- 6' Thus, D' is dense. Consequently, there is $n < \omega$ such that $g(n) \in D'$.
- 7' $g(n)$ witnesses $D \cap G \neq \emptyset$.

□

- 12° Thus, $\mathfrak{M} := \mathbf{Ult}(V, G)$ is well-founded. Let $j : V \prec \mathfrak{M}$ be the ultrapower embedding.
- 13° Let $\gamma := [\text{id}] \in \mathbf{Ord}$. We have $\gamma \in \bigcup_{i < \omega} j(g(i))$.
- 14° Let $\mathbb{Q} := \{p \in j(\mathbb{P}) : \gamma \in \bigcup_{i < |p|} p(i)\}$. We have $\mathbb{Q} \in \mathfrak{M}$ and it is a tree.

- 15° $(j(g(i)) : i < \omega)$ is a branch through $\mathbb{Q}$ in $V[g]$. By absoluteness of well-foundedness, there is a branch $h \in [\mathbb{Q}] \cap \mathfrak{M}$.
- 16° Inside $\mathfrak{M}$, we have $h \in [\mathbb{Q}] \subseteq [j(\mathbb{P})]$, implying that h is a play of $j(\text{PrecGame}(I))$ played according to $j(\sigma)$. In particular, $\bigcap_{i < \omega} h(i) = \emptyset$.
- 17° On the other hand, $\gamma \in \bigcap_{n < \omega} h(i)$, which is a contradiction.

□

1.2 Pre-saturation

- I. **Definition.** A normal ideal I on κ is *pre-saturated* if for every $G \rightsquigarrow V^{\mathbb{P}_I}$, model $\text{Ult}(V, G)$ is $<\kappa$ -closed.
- II. **Proposition.** Every pre-saturated ideal is precipitous.

Proof. Obvious. □

- III. Let I be ideal on κ and let $D \subseteq \mathbb{P}_I$ be pre-dense. For $S \in \mathbb{P}_I$, we define

$$D \restriction S := \{T \cap S : T \in D, T \parallel S\};$$

this set is clearly pre-dense below S .

- IV. **Proposition.** A normal ideal on I on κ is pre-saturated if and only if for every family $(A_\alpha : \alpha < \lambda)$ of anti-chains in $\mathbb{P}_I$, with $\lambda < \kappa$, and every $S_0 \in \mathbb{P}_I$, there exists $S \leq_{\mathbb{P}_I} S_0$ such that $|A_\alpha \restriction S| \leq \kappa$ for every $\alpha < \lambda$.

Proof.

Implication ($\Rightarrow$)

- 1° Let $(A_\alpha : \alpha < \lambda)$ be a family of maximal anti-chains in $\mathbb{P}_I$ ($\lambda < \kappa$) and let $S_0 \in \mathbb{P}_I$.
- 2° Let $\dot{f} : \lambda \rightarrow V$ in $V^{\mathbb{P}_I}$ such that: for every $G \rightsquigarrow V^{\mathbb{P}_I}$, $\dot{f}^G(\alpha) = j_G(T)$ where T is the unique element of $G \cap A_\alpha$ ($\alpha < \lambda$).
- 3° There is $S \leq S_0$ and $g : \kappa \rightarrow (\lambda \rightarrow V)$ such that $S \Vdash \dot{f} = [g]$. Let us show that S is as required.
- 4° Suppose that $T \in A_\alpha$ such that $T \parallel S$. Let $G \rightsquigarrow V^{\mathbb{P}_I}$ such that $G \ni T \cap S$.
- 5° Then $[g](\alpha) = j_G(T)$, so $\{\xi < \kappa : g(\xi)(\alpha) = T\} \in G$.
- 6° In particular, $T \in \{g(\xi)(\alpha) : \xi < \kappa, \alpha < \lambda\}$.
- 7° Thus, we have verified

$$\{T \in A_\alpha : T \parallel S\} \subseteq \{g(\xi)(\alpha) : \xi < \kappa, \alpha < \lambda\},$$

yielding $|A_\alpha \restriction S| \leq |\kappa \times \lambda| \leq \kappa$, for every $\alpha < \lambda$.

Implication ($\Leftarrow$)

8° Suppose that $\Vdash \dot{f} : \lambda \rightarrow V$ for some $\lambda < \kappa$. For every $S_0 \in \mathbb{P}_I$, we want to find some $S \leq S_0$ and a function $g : S \rightarrow V$ such that

$$S \Vdash [g] : \lambda \rightarrow \text{Ult}(V, \mathbb{G}),$$

$$(\forall \alpha < \lambda)(\Vdash [g](\alpha) = \dot{f}(\alpha)).$$

9° Let A_α be a maximal anti-chain whose elements decide the value of $\dot{\alpha}$ ($\alpha < \lambda$).

10° Let $S \leq S_0$ be such that $|A_\alpha \restriction S| \leq \kappa$.

11° Let us enumerate $A_\alpha \restriction S = \{T_\xi^\alpha : \xi < \kappa\}$ ($\alpha < \lambda$).

12° Up to replacing T_ξ^α with $T_\xi^\alpha - \bigcup_{\eta < \xi} T_\eta^\alpha$, we may assume that $T_\xi^\alpha \cap T_\zeta^\alpha = \emptyset$ for $\xi \neq \zeta$.

13° Let us define $g : \kappa \rightarrow (\lambda \rightarrow V)$ by setting

$$g(\xi)(\alpha) := \begin{cases} x, & \xi \in T_\eta^\alpha, T_\eta^\alpha \Vdash \dot{f}(\alpha) = x \\ 0, & \xi \in S - \bigcup_{\eta < \kappa} T_\eta^\alpha. \end{cases}$$

14° $\Vdash [g] : \lambda \rightarrow V$

15° For every $\alpha < \lambda$, we have $T_\eta^\alpha \Vdash [g](\alpha) = \dot{f}(\alpha)$ ($\eta < \kappa$), so $S \Vdash [g](\alpha) = \dot{f}(\alpha)$.

□

V. **Lemma.** Let I be a normal ideal on a successor κ . Then $\mathbb{P}_I$ collapses κ .

Proof.

1° Suppose that $\kappa = \lambda^+$.

2° Let $G \rightsquigarrow V^{\mathbb{P}_I}$ be arbitrary.

3° Then $\lambda < \kappa < j_G(\kappa) = j_G(\lambda^{+V}) = j_G(\lambda)^{+\text{Ult}(V, G)} = \lambda^{+\text{Ult}(V, G)}$.

4° This shows that there is a surjection $f : \lambda \rightarrow \kappa$ in $\text{Ult}(V, G)$.

5° Since $f \in \text{wfp Ult}(V, G)$, we conclude that $V[G] \models |\kappa| \leq \lambda < \kappa$.

□

VI. **Proposition.** Let I be a pre-saturated ideal on κ . Then $\mathbb{P}_I$ preserves κ^{+V} as a cardinal.

Proof.

1° Assume otherwise.

2° Let $G \rightsquigarrow V^{\mathbb{P}_I}$ be such that $|\kappa^{+V}|^{V[G]} < \kappa^{+V}$.

- 3° Since κ is not a cardinal in $V[G]$, there is $\lambda < \kappa$ and a surjection $f : \lambda \rightarrow \kappa^{+V}$.
- 4° Let $\dot{f}$ be a name for this function and let A_α be a maximal anti-chain whose each element T decide $\dot{f}(\alpha)$ as some $\gamma_{\alpha,T} < \kappa^{+V}$ ($\alpha < \lambda$).
- 5° Let $S_0 \in \mathbb{P}_I$ be arbitrary.
- 6° There is $S \leq S_0$ such that $|A_\alpha \restriction S| \leq \kappa$ for every $\alpha < \lambda$.
- 7° Let $\eta := \sup\{\gamma_{\alpha,T} : \alpha < \lambda, T \in A_\alpha, T \parallel A_\alpha\} < \kappa^{+V}$.
- 8° $S \Vdash \text{im } \dot{f} \subseteq \eta + 1 < \kappa^{+V}$.
- 9° Since we have established density, we conclude $\Vdash \text{im } \dot{f} \neq \kappa^{+V}$, which is a contradiction.

□

VII. Proposition. Let I be a pre-saturated ideal on successor κ and let $G \rightsquigarrow V^{\mathbb{P}_I}$. Then $j_G(\kappa) = \kappa^{+V}$.

Proof.

- 1° Let λ be such that $\kappa = \lambda^{+V}$.
- 2° Then $j_G(\kappa) = \lambda^{+\text{Ult}(V,G)}$.
- 3° Since there are no cardinals between λ and κ^{+V} in $V[G]$, since $\text{Ult}(V,G)$ is λ -closed, and Since κ^{+V} is a cardinal in $V[G]$, we conclude $\lambda^{+\text{Ult}(V,G)} = \kappa^{+V}$.
- 4° Thus, $j_G(\kappa) = \kappa^{+V}$.

□

1.3 Saturation

- I. **Remark.** It is well known that for a normal ideal I on κ , the poset $\mathbb{P}_I$ is not κ -c.c. (use Ulam's matrix).
- II. **Definition.** A normal ideal I on κ is *saturated* if $\mathbb{P}_I$ is κ^+ -c.c.
- III. **Remark.** If I is κ -complete, we have that $P(\kappa)/I$ is κ -complete. If I is also normal, $P(\kappa)/I$ is in fact κ^+ -complete:

$$\sum_{\alpha < \kappa} [A_\alpha] = [\bigvee_{\alpha < \kappa} A_\alpha], \quad \prod_{\alpha < \kappa} [A_\alpha] = [\bigwedge_{\alpha < \kappa} A_\alpha].$$

In the case of saturation, we additionally have that $P(\kappa)/I$ is κ^+ -c.c., so it is complete. In particular, $\mathbb{B}(\mathbb{P}_I) = P(\kappa)/I$.

IV. **Proposition.** Let I be a normal ideal on κ . Then I is saturated if and only if for every $\tau \in V^{\mathbb{P}_I}$ such that

$$\Vdash \tau : \kappa \rightarrow V \wedge \tau \in V,$$

there exists $f : \kappa \rightarrow V$ such that

$$\Vdash [\tau] = [f].$$

Proof.

Implication ($\Rightarrow$)

- 1° Let D be the set of all $S \in \mathbb{P}_I$ such that S decides the value of τ as some $f_S : \omega_1 \rightarrow V$.
- 2° D is dense and open.
- 3° Let $A \subseteq D$ be a maximal anti-chain. $|A| \leq \omega_1$.
- 4° Let $(S_\alpha : \alpha < \gamma)$ be an injective enumeration of A ($\gamma \leq \omega_1$).
- 5° $T_\alpha := S_\alpha - \bigcup_{\beta < \alpha} S_\beta \in \mathbb{P}_I$ are disjoint and predense, for $\alpha < \gamma$.
- 6° Let $f : \omega_1 \rightarrow V$ be defined by

$$f(\alpha) := \begin{cases} f_{S_\xi}(\alpha), & \alpha \in T_\xi, \xi < \gamma, \\ 0, & \text{otherwise.} \end{cases}$$

- 7° $T_\alpha \Vdash [f] = [f_{S_\alpha}] = [\tau]$
- 8° Since T_α 's are predense, we have $\Vdash [\tau] = [f]$.

Implication ($\Leftarrow$)

- 9° Let $A \subseteq \mathbb{P}_I$ be a maximal anti-chain. Let τ be a name for a function $\kappa \rightarrow V$ in $V^{\mathbb{P}_I}$ such that for every $T \in A$,

$$T \Vdash \tau \equiv T.$$

- 10° Since $\Vdash \tau \in V$, there is $f : \kappa \rightarrow V$ such that $\Vdash [\tau] = [f]$.
- 11° For every $T \in A$, we have $T \Vdash [f] \equiv T$. In particular, $T \in \text{im } f$.
- 12° Thus, $A \subseteq \text{im } f$, and consequently $|A| \leq \kappa$.

□

V. **Proposition.** A saturated ideal is pre-saturated.

Proof.

- 1° Let I be a saturated ideal on κ and let $(\tau_\alpha : \alpha < \lambda)$ be an arbitrary sequence of names such that

$$\Vdash \tau_\alpha \in (\kappa \rightarrow V) \cap V$$

for every $\alpha < \lambda < \kappa$.

2° We want to find τ such that

$$\Vdash \tau \in (\kappa \rightarrow (\lambda \rightarrow V)) \cap V,$$

$$(\forall \alpha < \lambda)(\Vdash [\tau](\alpha) = \tau_\alpha).$$

3° For $\alpha < \lambda$, there is $f_\alpha : \kappa \rightarrow V$ such that $\Vdash [\tau_\alpha] = [f_\alpha]$.

4° Let $f : \kappa \rightarrow (\lambda \rightarrow V)$ be defined by

$$f(\xi)(\alpha) := f_\alpha(\xi) \quad (\xi < \kappa, \alpha < \lambda).$$

5° For $\alpha < \lambda$,

$$\Vdash [f](\alpha) = [(f(\xi)(\alpha) : \xi < \kappa)] = [(f_\alpha(\xi) : \xi < \kappa)] = [f_\alpha].$$

□

VI. **Remark.** We could have proved the previous proposition more readily by simply referencing [1-IV](#).

2 The Non-stationary Ideal

2.1 Precipitousness

- I. In this section, we study the regularity properties in the case of the non-stationary ideal $\text{NS} = \text{NS}_{\omega_1}$ on ω_1 . We study with the well-foundedness of the generic ultrapower.
- II. **Definition.** Let $\theta > 2^{\omega_1}$ be regular, let $D \subseteq \mathbb{P}_{\text{NS}}$ be pre-dense, and let $M \prec (H_\theta, \in, D)$ be countable. M is said to *capture* D if there is $S \in D \cap M$ such that $M \cap \omega_1 \in S$.
- III. **Notation.** If $k : M \prec N$ is an elementary embedding between models of set theory, we denote by U_k the M -ultrafilter derived from k at $\kappa := \text{crit } k$:

$$(\forall X \in P(\kappa) \cap M)(X \in U_k \iff \kappa \in k(X)).$$

Recall that there is a canonical realizability embedding k^* defined by $k^*([f]_{U_k}) = k(f)(\kappa)$ (for $f \in (\kappa \rightarrow M) \cap M$).

$$\begin{array}{ccc} N & & \\ \uparrow k & \nwarrow k^* & \\ M & \xrightarrow{j_{U_k}^M} & \text{Ult}(M, U_k) \end{array}$$

- IV. **Remark.** Let $\theta > 2^{\omega_1}$ be regular, let $D \subseteq \mathbb{P}_{\text{NS}}$ be pre-dense, let $M \prec (H_\theta, \in, D)$ be countable, and let $k : \overline{M} \cong M \prec H_\theta$ be the anti-collapse. We see that M captures D if and only if $D \cap U_k \neq \emptyset$.
- V. **Definition.** Let $\theta > 2^{\omega_1}$ be regular and let $M \prec H_\theta$ be countable. Model M is *self-generic* if it captures every pre-dense $D \subseteq \mathbb{P}_{\text{NS}}$ that it contains.
- VI. **Remark.** If $k : \overline{M} \cong M \prec H_\theta$ is the anti-collapse, we see that M is self-generic if and only if $U_k \rightsquigarrow \overline{M}^{\mathbb{P}_{\text{NS}}}$.
- VII. **Definition.** $\text{selfGen} \subseteq [H((2^{\omega_1})^+)]^\omega$ is defined as the set of all self-generic $M \prec H((2^{\omega_1})^+)$.
- VIII. **Theorem.** NS_{ω_1} is precipitous if and only if selfGen is projectively satationary².

Proof.

Implication ($\Rightarrow$)

- 1° Let us denote $\theta := (2^{\omega_1})^+$ and let $S \in \mathbb{P}_{\text{NS}}$ be arbitrary. We want to show that

$$a := \{M \in [H_\theta]^\omega : M \cap \omega_1 \in S \wedge M \text{ self-generic}\}$$

is stationary.

- 2° Let $\mathbb{A} = (H_\theta, \in, R, \dots)$ be an arbitrary structure on a countable language, where R is a well-order. We want to find $M \prec \mathbb{A}$ such that $M \in a$.
- 3° Let $G \ni S$ be $V^{\mathbb{P}_{\text{NS}}}$ -generic.
- 4° $j := j_G : V \prec \text{Ult}(V, G) =: \mathfrak{M}$
- 5° We will be done if in $\mathfrak{M}$, we find a countable M such that:
- a. $M \prec j(\mathbb{A})$
 - b. $M \cap \omega_1^{\mathfrak{M}} \in j(S)$
 - c. M is self-generic.
- 6° Let $(\text{fst}, \text{snd}) \in \mathfrak{M}$ be a surjection from ω onto pairs (h, s) where h is a Skolem function for $j(\mathbb{A})$ and $s \in \omega^{\text{ar } h}$. We can make sure that for every $n < \omega$, we have $\max \text{snd}(n) \leq 3n$.
- 7° Let $\text{list} : \omega \rightarrow \omega_1^V$ be a bijection, with $\text{list} \in \mathfrak{M}$.
- 8° Let T be the tree on $j(H_\theta^V)$ of attempts to construct a sequence x satisfying for every $n < \omega$:
- a. $x(3n) := \text{list}(n)$

²I.e. it is stationary and it is compatible (in the poset of stationary subsets of $[H((2^{\omega_1})^+)]^\omega$ ordered by inclusion) with all the sets of the form

$$\{M \subseteq [H((2^{\omega_1})^+)]^\omega : M \cap \omega_1 \in S\},$$

where S is stationary in ω_1

- b. denote $h := \text{fst}(n)$;
- c. denote $s := \text{snd}(n)$;
- d. denote $k := |s| = \text{ar } h$;
- e. $x(3n+1) := h(x(s(0)), \dots, x(s(k-1)))$ if this is not in $\omega_1^M - \omega_1^V$, otherwise halt;
- f. for $x(3n+2)$, we have three cases:
 - i. if $x(3n+1)$ is a predense subset of $\mathbb{P}_{\text{NS}}^M$ and there is some $B \in x(3n+1)$ with $\omega_1^V \in B$, we take one such B for $x(3n+2)$;
 - ii. if $x(3n+1)$ is a predense subset of $\mathbb{P}_{\text{NS}}^M$ and there is no $B \in x(3n+1)$ with $\omega_1^V \in B$, we halt;
 - iii. if $x(3n+1)$ is not a predense subset of $\mathbb{P}_{\text{NS}}^M$, we take $x(3n+2) := 0$.

9° Note that $T \in \mathfrak{M}$.

10° A branch through T enumerates a countable model M with the following properties:

- a. $M \prec j(\mathbb{A})$;
- b. $M \cap \omega_1^{\mathfrak{M}} = \omega_1^V$;
- c. for every $D \in M$ predense in $\mathbb{P}_{\text{NS}}^{\mathfrak{M}}$, we have $B \in D \cap M$ with $\omega_1^V \in B$.

Property **b** ensures $M \cap \omega_1^{\mathfrak{M}} \in j(S)$, while **b** and **c** ensure that M is self-generic.

11° In other words, we are done if we show that T has a branch in M .

12° **Claim.** T has a branch in $V[G]$.

Proof.

- 1' $j[H_\theta^V] \prec j(\mathbb{A})$. Let $\mathbb{B}$ denote the induced structure.
- 2' **Subclaim.** For every $D \in j[H_\theta^V]$ predense in $\mathbb{P}_{\text{NS}}^{\mathfrak{M}}$, there is some $B \in D$ with $B \in j[H_\theta^V]$ and $\omega_1^V \in B$.

Proof.

- 1'' $D = j(D')$ for some predense $D' \subseteq \mathbb{P}_{\text{NS}}^V$.
- 2'' There is some $B' \in D' \cap G$.
- 3'' Let $B := j(B')$.
- 4'' B is as required.

□

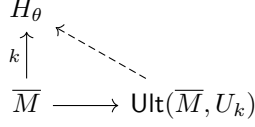
- 3' Let $f : j[H_\theta^V] \rightarrow j[H_\theta^V]$ satisfy: if $D \in j[H_\theta^V]$ is predense in $\mathbb{P}_{\text{NS}}^{\mathfrak{M}}$, then $f(D) \in D \cap j[H_\theta^V]$ and $\omega_1^V \in f(D)$; otherwise, $f(D) = 0$.
- 4' Let $N := \text{Hull}(\omega_1^V, (\mathbb{B}, f))$.
- 5' We have $N \prec \mathbb{B} \prec j(\mathbb{A})$ and N is closed for f .
- 6' By enumerating N in the appropriate manner, we get a branch through T .

□

13° The absoluteness of well-foundedness now yields the conclusion.

Implication ($\Leftarrow$)

- 14° Assume otherwise, that **NS** is not precipitous.
- 15° Then there is regular θ large enough and $S \in \mathbb{P}_{\text{NS}}$ so that $S \Vdash \ulcorner \text{Ult}(H_\theta^V, \mathbb{G}) \text{ is ill-founded} \urcorner$.
- 16° Then there is a countable hull $k : (\overline{M}, \in, \overline{S}) \prec (H_\theta, \in, S)$ which is self-generic and such that $\text{crit } k \in S$.
- 17° $S \in U_k$ and $U_k \rightsquigarrow \overline{M}^{\mathbb{P}_{\text{NS}}}$.
- 18° Thus, $\text{Ult}(\overline{M}, U_k)$ is ill-founded.
- 19° This is a contradiction, since the diagram



can be completed.

□

2.2 Saturation

- I. **Definition.** Let $\theta > 2^{\omega_1}$ be regular and let $D \subseteq \mathbb{P}_{\text{NS}}$ be pre-dense. Set $\text{capt}(D, \theta) \subseteq [H_\theta]^\omega$ is defined as the set of all countable $M \prec H_\theta$ that capture D .
- II. **Remark.** Family $(S_\alpha : \alpha < \omega_1)$ of stationary sets is pre-dense if and only if $\nabla_{\alpha < \omega_1} S_\alpha$ contains a club.
- III. **Proposition.** Let $\theta > 2^{\omega_1}$ be regular. A pre-dense set $D \subseteq \mathbb{P}_{\text{NS}}$ contains a pre-dense subset of cardinality $\leq \omega_1$ if and only if $\text{capt}(D, \theta)$ is club.

Proof.

Implication ($\Rightarrow$)

- 1° Let $j : (M, \in, \overline{D}) \prec (H_\theta, \in, D)$ be an arbitrary countable hull. We will show that $\text{im } j \in \text{capt}(D, \theta)$. This will suffice for the conclusion.
- 2° We aim to show that there is $S \in \overline{S}$ such that $\omega_1^M \in j(S)$.
- 3° $H_\theta \models (\exists f : \omega_1 \rightarrow D)(\nabla f \in \text{club})$
- 4° Thus, $M \models (\exists f : \omega_1 \rightarrow \overline{D})(\nabla f \in \text{club})$.
- 5° Let $f : \omega_1^M \rightarrow \overline{D}$ with $f \in M$ and $\nabla f \in \text{club}^M$.
- 6° Then $j(f) : \omega_1^M \rightarrow D$, with $\nabla j(f) \in \text{club}$ and $j(f) \restriction \omega_1^M = f$.
- 7° $\nabla j(f) \cap \omega_1^M = \nabla f$ is unbounded in ω_1^M .
- 8° Since $\nabla j(f)$ is a club, we conclude $\omega_1^M \in \nabla j(f)$.

9° Thus, there is $\alpha < \omega_1^M$ such that $\omega_1^M \in j(f)(\alpha) = j(f(\alpha))$.

10° In other words, $S := f(\alpha) \in \overline{D}$ is as required.

Implication ($\Leftarrow$)

11° Let D be a predense. We will find $(S_\alpha)_\alpha \in D^{\omega_1}$ such that $\nabla_{\alpha < \omega_1} S_\alpha$ contains a club, which will suffice for the conclusion.

12° **Claim.** For every regular θ' large enough, there exists a parameter $p \in H_{\theta'}$ such that for every countable $N \prec (H_{\theta'}, \in, p)$, we have $N \in \text{capt}(D, \theta)$.

Proof.

1' Since $\text{capt}(D, \theta)$ is a club, there is $F : H_\theta^{<\omega} \rightarrow H_\theta$ such that $C_F \subseteq \text{capt}(D, \theta)$.

2' Let $\theta' > 2^\theta$ be arbitrary regular cardinal.

3' $p := (F, \theta) \in H_\theta$

4' For every countable $N \prec (H_{\theta'}, \in, p)$, we have that $N \cap H(\theta) \prec H(\theta)$ and it is closed for F (since $F \in N$).

5' Thus, $N \cap H(\theta) \in \text{capt}(D, \theta)$.

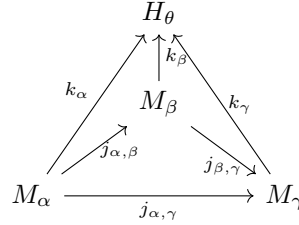
6' But then also $N \in \text{capt}(D, \theta')$.

□

13° We may assume w.l.o.g. that θ witnesses the claim. Let $p \in H_\theta$ be the appropriate parameter.

14° Let $k_0 : (M_0, \in, \bar{p}, \overline{D}, \kappa_0) \prec (H_\theta, \in, p, D, \omega_1)$ be arbitrary.

15° By induction, we define models M_α , embeddings $j_{\alpha, \beta} : M_\alpha \prec M_\beta$, and embeddings $k_\alpha : M_\alpha \prec H_\theta$ such that diagrams



commute, for $\alpha \leq \beta \leq \gamma \leq \omega_1$.

16° The base step is handled, while the limit step is done by passing to the direct limits.

17° At the successor step, we first derive the measure U_{k_α} from k_α at $\kappa_\alpha := j_{0, \alpha}(\kappa_0) = \omega_1^{M_\alpha}$. Then we set $M_{\alpha+1} := \text{Ult}(M_\alpha, U_{k_\alpha})$ and take $j_{\alpha, \alpha+1}$ to be the ultrapower embedding. Embedding $k_{\alpha+1}$ is the canonical realizability embedding ($k_{\alpha+1}([f]_{U_{k_\alpha}}) = k_\alpha(f)(\kappa_\alpha)$). Other embeddings are now unambiguous.

18° We note that $\{\kappa_\alpha : \alpha < \omega_1\}$ is a club in ω_1 .

- 19° Let $(S_\alpha : \alpha < \omega_1)$ have the property that $j_{0,\alpha}(\overline{D}) = \{k_\alpha^{-1}(S_\xi) : \xi < \omega\alpha\}$.
- 20° $C := \{\alpha < \omega_1 : \kappa_\alpha = \omega\alpha = \alpha\}$ is a club.
- 21° Let us verify that $C \subseteq \nabla_{\alpha < \omega_1} S_\alpha$.
- 22° Let $\alpha \in C$ be arbitrary.
- 23° Since $p \in \text{im } k_0 \subseteq \text{im } k_\alpha$, we have $\text{im } k_\alpha \in \text{capt}(D, \theta)$.
- 24° Then there is $S \in k_\alpha^{-1}(D)$ such that $\kappa_\alpha \in k_\alpha(S)$.
- 25° Hence, there is $\xi < \omega\alpha = \alpha$ such that $\kappa_\alpha \in S_\xi$.
- 26° This means $\alpha = \kappa_\alpha \in S_\xi$ for this $\xi < \alpha$.
- 27° We conclude $\alpha \in \nabla_{\alpha < \omega_1} S_\xi$.

□

IV. **Proposition.** $\mathbb{P}_{\text{NS}}$ is saturated if and only if for every regular θ large enough, there are club many $M \prec H_\theta$ that are self-generic.

Proof.

Implication ($\Rightarrow$)

- 1° If $\mathbb{P}_{\text{NS}}$ is saturated, then by 2-III, for every predense $D \subseteq \mathbb{P}_{\text{NS}}$, set $\text{capt}(D, \theta)$ is club.
- 2° By enlarging θ if necessary, we can find $p_D \in H_\theta$ such that

$$M \prec H_\theta, |M| = \omega, p_D \in M \implies M \in \text{capt}(D, \theta).$$

- 3° We then have $p := (p_D : D \subseteq \mathbb{P}_{\text{NS}} \text{ predense}) \in H_\theta$ and for every countable $M \prec H_\theta$ with $p \in M$, we have that M is self-generic.

Implication ($\Leftarrow$)

- 4° If there are club many self-generic, that club is contained in $\text{capt}(D, \theta)$ for every predense $D \subseteq \mathbb{P}_{\text{NS}}$.

□

3 Models for the Regularity Properties

- I. **Lemma.** Let M be an inner model, let $\mathbb{P}$ be a poset, let $p \in \mathbb{P}$, let $\dot{x} \in M^\mathbb{P}$, and let $\phi(\dot{x})$ be a formula. Suppose that $(M, \mathbb{P}, p) \Vdash \phi(\dot{x})$. Then $(V, \mathbb{P}, p) \Vdash \phi^{M[\dot{g}]}(\dot{x})$.

Proof.

- 1° Let $g \rightsquigarrow V^\mathbb{P}$ with $p \in g$ be arbitrary. We want to show that $M[g] \models \phi[\dot{x}^g]$.
- 2° It holds $g \rightsquigarrow M^\mathbb{P}$.

3° Consequently, $M[g] \models \phi[\dot{x}^g]$.

□

II. **Lemma.** Let M be an inner model, let $\mathbb{P}$ be a poset, let $p \in \mathbb{P}$, let $\dot{x} \in M^{\mathbb{P}}$, and let $\phi(\dot{x})$ be a formula. Then

$$(M, \mathbb{P}, p) \Vdash \phi(\dot{x}) \iff (V, \mathbb{P}, p) \Vdash \phi^{M[\dot{g}]}(\dot{x}).$$

Proof.

- 1° Implication $(\Rightarrow)$ is verified in the previous lemma.
- 2° Assume now $(M, \mathbb{P}, p) \nVdash \phi(\dot{x})$ and let us verify $(V, \mathbb{P}, p) \nVdash \phi^{M[\dot{g}]}(\dot{x})$.
- 3° There is $q \leq p$ such that $(M, \mathbb{P}, q) \Vdash \phi(\dot{x})$.
- 4° By the previous lemma, it holds that $(V, \mathbb{P}, q) \Vdash \phi^{M[\dot{g}]}(\dot{x})$.
- 5° Thus, $(V, \mathbb{P}, p) \nVdash \phi^{M[\dot{g}]}(\dot{x})$.

□

III. **Proposition.** Let κ be a regular cardinal, I be a normal ideal on κ , let $\mathbb{P}$ be a κ -c.c. poset, let $g \rightsquigarrow V^{\mathbb{P}}$, and let J be the ideal on κ in $V[g]$ generated by I . Then J is normal in $V[G]$.

Proof. This follows by the proof of [2, Proposition 2-VI].

□

IV. **Theorem.** Let κ be measurable and let μ be a normal measure on κ . Let $g \rightsquigarrow V^{\text{Col}(\omega, < \kappa)}$ and let I be the ideal dual to the filter generated by μ in $V[G]$. Then I is normal and precipitous.

Proof.

- 1° Normality follows from the previous lemma. We verify precipitousness.
- 2° It suffices to show that for every $S \in \mathbb{P}_I$, there exists $G \rightsquigarrow V[g]^{\mathbb{P}_I}$ such that $\text{Ult}(V[g], G)$ is well-founded.
- 3° Note that $S \in I$ if and only if there exists $Z \in \mu$ such that $S \cap Z = \emptyset$.
- 4° Let $j := j_\mu : V \prec \text{Ult}(V, \mu) =: \mathfrak{M}$.
- 5° Let $\dot{S} \in V^{\text{Col}(\omega, < \kappa)}$ be an arbitrary name for a subset of κ and let us denote $\tau_{\dot{S}} := j(\dot{S}) \in \mathfrak{M}^{\text{Col}(\omega, < j(\kappa))}$.

$$\dot{S}^g \in I \tag{1}$$

$$\iff (\exists Z \in \mu)(\dot{S}^g \cap Z = \emptyset) \tag{2}$$

$$\iff (\exists Z \in \mu)(\exists p \in g)(p \Vdash \dot{S} \cap Z = \emptyset) \tag{3}$$

$$\iff (\exists p \in g)(\exists Z \in \mu)(\forall \alpha \in Z)(p \Vdash \alpha \notin \dot{S}) \tag{4}$$

$$\iff (\exists p \in g)(\{\alpha < \kappa : p \Vdash \alpha \notin \dot{S}\} \in \mu) \tag{5}$$

$$\iff (\exists p \in g)((\mathfrak{M}, \text{Col}(\omega, < j(\kappa)), p) \Vdash \kappa \notin \tau_{\dot{S}}) \tag{6}$$

$$\iff (\exists p \in g)(\mathfrak{M} \models (p \Vdash_{\text{Col}(\omega, < \kappa)} \emptyset \Vdash_{\text{Col}(\omega, [\kappa, j(\kappa)))} \kappa \notin \tau_{\dot{S}}^{g \times \dot{h}})) \tag{7}$$

$$\iff \mathfrak{M}[g] \models (\Vdash_{\text{Col}(\omega, [\kappa, j(\kappa)))} \kappa \notin \tau_{\dot{S}}^{g \times \dot{h}}). \tag{8}$$

6° Let $S_0 \in \mathbb{P}_I$ be arbitrary. We want to find $G \rightsquigarrow V[g]^{\text{Col}(\omega, [\kappa, j(\kappa)])}$ such that $G \ni S_0$ and that $\text{Ult}(V[g], G)$ is well-founded.

7° Let $\dot{S}_0 \in V^{\text{Col}(\omega, < \kappa)}$ be such that $\dot{S}_0^g = S_0$. Since $\dot{S}_0^g \notin I$, it holds

$$\mathfrak{M}[g] \models (\Vdash_{\text{Col}(\omega, [\kappa, j(\kappa)])} \kappa \notin \tau_{\dot{S}_0}^{g \times h}).$$

8° Hence, there exists $h \rightsquigarrow V[g]^{\text{Col}(\omega, [\kappa, j(\kappa)])}$ such that $\mathfrak{M}[g][h] \models (\kappa \in \tau_{\dot{S}_0}^{g \times h})$.

9° Mapping

$$k : V[g] \prec \mathfrak{M}[g][h] : \tau^g \mapsto j(\tau)^{g \times h}$$

is correctly defined, elementary, and satisfies $k \restriction V = j$. In particular, $\text{crit}(k) = \kappa = \omega_1^{V[g]}$ and $k(\dot{S}^g) = \tau_{\dot{S}}^{g \times h}$ for $\dot{S} \in V^{\text{Col}(\omega, < \kappa)}$ a name for a subset of κ .

10° Let G be the measure on κ derived from k at κ . In other words, for a name $\dot{S} \in V^{\text{Col}(\omega, < \kappa)}$ for a subset of κ , we have

$$\dot{S}^g \in G \iff \kappa \in \tau_{\dot{S}}^{g \times h}.$$

11° Since diagram

$$\begin{array}{ccc} & \mathfrak{M}[g][h] & \\ k \uparrow & \swarrow & \\ V[g] & \xrightarrow{j_G^{V[g]}} & \text{Ult}(V[g], G) \end{array}$$

can be completed by the canonical realizability embedding, we conclude that $\text{Ult}(V[g], G)$ is well-founded. By 8°, we also have $S_0 \in G$, so we will be done if we show $G \rightsquigarrow V[g]^{\mathbb{P}_I}$.

12° **Claim.** $G \rightsquigarrow V[g]^{\mathbb{P}_I}$

Proof.

1' **Subclaim.** $G \subseteq \mathbb{P}_I$

Proof.

1'' Let $S \in G$ be arbitrary and let us verify that $S \notin I$.

2'' Since $\kappa \in k(S)$, we have $\mathfrak{M}[g][h] \models \kappa \in \tau_{\dot{S}}^{g \times h}$.

3'' By 5°, we have $S \notin I$. □

2' Let D be dense in $\mathbb{P}_I$. We want to show that $D \cap G \neq \emptyset$.

3' Assume otherwise. We will reach a contradiction by showing that D is not dense.

4' There exists $p \in g \times h$ such that

$$(V, \text{Col}(\omega, < j(\kappa)), p) \Vdash \dot{D} \cap \dot{G} = \emptyset.$$

5' Let $f \in (\kappa \rightarrow \text{Col}(\omega, < \kappa)) \cap V$ be such that $p = [f]_\mu$.

6' Let $T := \{\alpha < \kappa : f(\alpha) \in g\}$.

7' **Subclaim.** $T \in \mathbb{P}_I$

Proof.

1'' For every $\alpha < \kappa$, it holds that

$$(V, \text{Col}(\omega, < \kappa), f(\alpha)) \Vdash \alpha \in \dot{T}.$$

2'' Hence,

$$(\mathfrak{M}, \text{Col}(\omega, < j(\kappa)), p) \Vdash \kappa \in \tau_{\dot{T}}.$$

3'' Seeing that $p \in g \times h$, we conclude $\mathfrak{M}[g][h] \models \kappa \in \tau_{\dot{T}}^{g \times h}$.

4'' By 5°, we obtain $T \notin I$.

□

8' We will show that there is no $S \in D$ with $S \subseteq T$, reaching the contradiction.

9' Let $S \in D$ be arbitrary. It suffices to verify that $S \cap T \in I$.

10' By 12°4', we have

$$(V, \text{Col}(\omega, < j(\kappa)), p) \Vdash \dot{S} \notin \dot{G},$$

which is equivalent to

$$(V, \text{Col}(\omega, < j(\kappa)), p) \Vdash \kappa \notin \tau_{\dot{S}}.$$

11' Lemma 3-II implies

$$(\mathfrak{M}, \text{Col}(\omega, < j(\kappa)), p) \Vdash \kappa \notin \tau_{\dot{S}}.$$

12' Recalling that $\tau_{\dot{S}} = j(\dot{S})$, we conclude by Łoś's theorem that

$$U := \{\alpha < \kappa : (V, \text{Col}(\omega, < \kappa), f(\alpha)) \Vdash \alpha \notin \dot{S}\} \in \mu.$$

13' We will be done with the argument if we show $S \cap T \cap U = \emptyset$.

14' **Subclaim.** $S \cap T \cap U = \emptyset$

Proof.

1'' Assume otherwise. Then there exists $\alpha \in S \cap T \cap U$.

2'' Since $\alpha \in U$, we have

$$(V, \text{Col}(\omega, < \kappa), f(\alpha)) \Vdash \alpha \notin \dot{S}.$$

3'' Since $\alpha \in T$, we have $f(\alpha) \in g$.

4'' These two statements together imply $\alpha \notin \dot{S}^g = S$. This is a contradiction.

□

□

□

V. **Theorem** (Shelah). Suppose δ is a Woodin cardinal. Then there exists a poset $\mathbb{P} \subseteq \delta$ such that

- a. $\mathbb{P}$ is semi-proper,
- b. $\mathbb{P}$ is δ -c.c.,
- c. NS_{ω_1} is saturated in $V^{\mathbb{P}}$.

Proof. See [3, Corollary [3.3.7]]. Its proof shows that we can find $\mathbb{P} \subseteq \delta$. □

VI. **Theorem.** Suppose MM holds. Then NS_{ω_1} is saturated.

Proof. See [1, Theorem 12]. □

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